TECHNICAL REPORT

Integrated Business Intelligence Analysis

End-to-End Business Data Analysis Capstone Project

1. Introduction

Data plays an important role in helping organizations understand their performance and make better business decisions. Companies collect large amounts of operational and customer data, but meaningful insights can only be obtained through systematic analysis. In this project, an end-to-end business data analysis was performed using three datasets related to sales performance, customer churn, and housing market prices. The objective was to apply practical data analytics techniques to explore patterns, identify key performance indicators, and generate insights that can support real-world business decisions. The project demonstrates how data analysis tools and visualization techniques can be used to convert raw datasets into understandable and actionable business insights.

2. Problem Statement

Organizations often face challenges such as:

- Uneven sales performance across regions
- Customer attrition affecting revenue stability
- Difficulty determining accurate pricing strategies

The project addresses these issues by applying analytical methods to identify patterns, evaluate performance indicators, and recommend data-driven solutions.

3. Objectives

The objectives of this project are:

1. Perform data cleaning and preprocessing.
2. Conduct exploratory data analysis (EDA).
3. Identify trends affecting business performance.
4. Calculate key performance indicators (KPIs).
5. Analyze customer churn behavior.
6. Examine housing price determinants.

7. Develop actionable business recommendations.
8. Create a reproducible analytics workflow.
9. Document the complete analysis process.
10. Present findings in a professional format.

4. Dataset Description

Three datasets were used in this analysis.

Dataset	Description	Records
sales_data.csv	Product sales transactions	100
house_prices.csv	Real estate property data	300
customer_churn.csv	Customer retention information	500

Total observations analyzed exceeded 900 records.

Key Attributes

Sales Dataset

- Date
- Product
- Quantity
- Price
- Region
- Total Sales

House Prices Dataset

- Area
- Bedrooms
- Bathrooms
- Property Type
- Location
- Price

Customer Churn Dataset

- Tenure
- Monthly Charges
- Contract Type
- Churn Indicator

5. Methodology

The project followed a structured analytical workflow similar to real-world data analysis projects. Each stage of the analysis was carried out step by step, starting from data loading and cleaning, followed by exploratory analysis, visualization, and business interpretation.Python and Jupyter Notebook were used to implement the workflow, allowing the analysis to remain reproducible and easy to understand.

5.1 Data Collection

Datasets were provided in CSV format and imported using Python Pandas.

5.2 Data Cleaning

Cleaning steps included:

- Data type conversion
- Missing value inspection
- Logical validation checks
- Feature engineering

A revenue category feature was created to classify sales levels.

5.3 Exploratory Data Analysis (EDA)

EDA was conducted to understand data distributions and detect patterns using:

- Histograms
- Bar charts
- Scatter plots
- Aggregated comparisons

5.4 Analytical Techniques Used

1. Descriptive Statistical Analysis
2. Comparative Analysis
3. Behavioral Pattern Analysis

6. Tools and Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Jupyter Notebook

These tools enabled efficient data manipulation, visualization, and analysis.

7. System Architecture

Raw Data

↓

Data Cleaning

↓

Exploratory Analysis

↓

Advanced Analysis

↓

Business Insights

↓

Recommendations

The workflow ensures reproducibility and structured analysis.

8. Exploratory Data Analysis Results

During exploratory analysis, patterns and variations within the datasets were examined to better understand relationships between variables before performing detailed analysis.

```python
sales.head()
house.head()
churn.head()
```

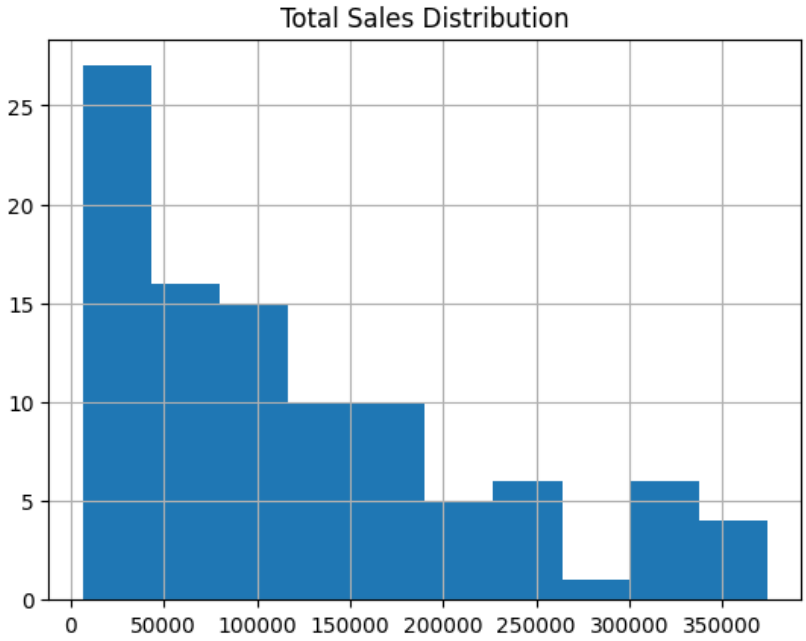
✓ 0.0s Python

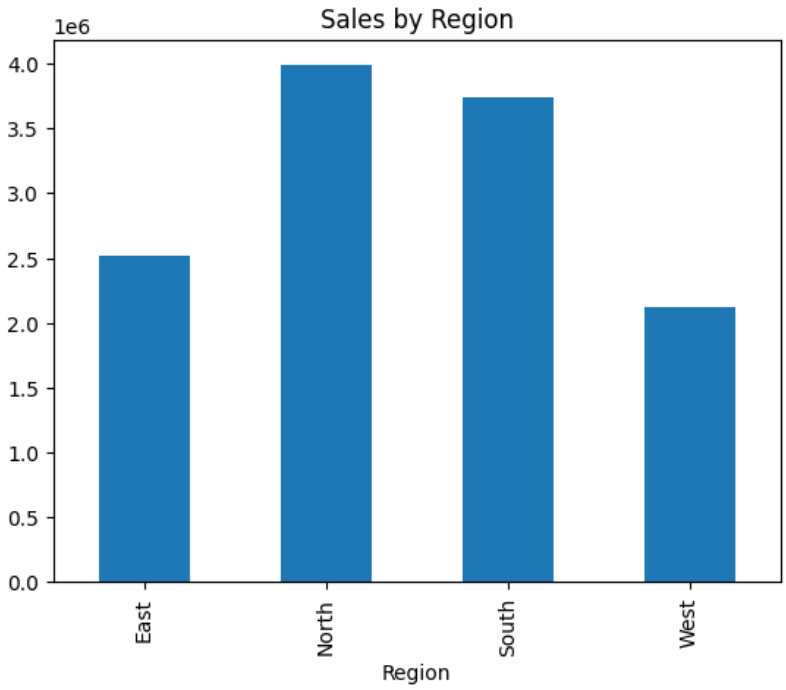
	CustomerID	Tenure	MonthlyCharges	TotalCharges	Contract	PaymentMethod	PaperlessBilling	SeniorCitizen	Churn
0	C00001	6	64	1540	One year	Credit Card	No	1	0
1	C00002	21	113	1753	Month-to-month	Electronic Check	Yes	1	0
2	C00003	27	31	1455	Two year	Credit Card	No	1	0
3	C00004	53	29	7150	Month-to-month	Electronic Check	No	1	0
4	C00005	16	185	1023	One year	Electronic Check	No	1	0

8.1 Sales Analysis

Sales distribution analysis revealed:

- Majority of transactions fall within mid-revenue ranges.
- Regional variations significantly affect revenue totals.
- Certain product categories contribute higher revenue.

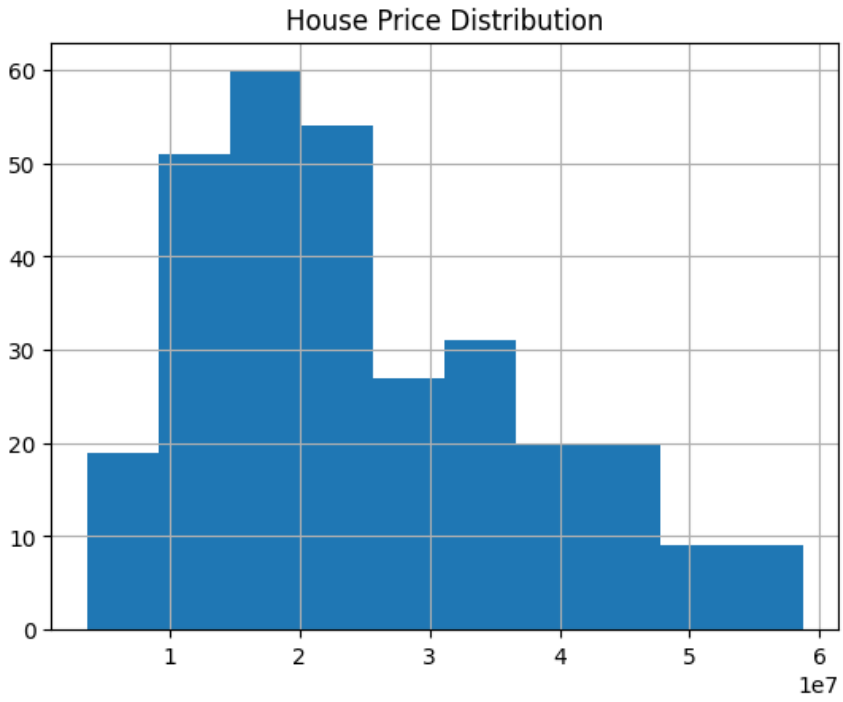




8.2 Housing Market Analysis

Key observations include:

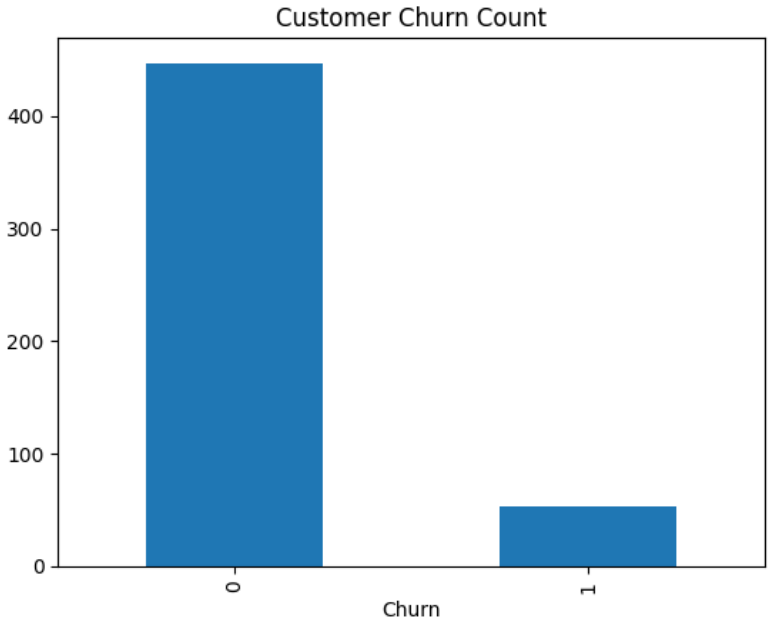
- Property area strongly correlates with price.
- Houses dominate market listings.
- Mid-price properties show higher frequency.



8.3 Customer Churn Analysis

Findings indicate:

- Overall churn rate approximately 10%.
- Customers with longer tenure demonstrate higher retention.
- Contract duration influences churn likelihood.



9. Key Performance Indicators (KPIs)

The following KPIs were calculated:

- Total Sales Revenue
- Average Property Price
- Customer Churn Rate

These metrics provide measurable indicators of business performance.

```
total_revenue = sales["Total_Sales"].sum()
avg_house_price = house["Price"].mean()

total_revenue, avg_house_price
```

✓ 0.0s Python

```
(np.int64(12365048), np.float64(24883658.333333332))
```

10. Validation and Testing

Validation checks were implemented to ensure data reliability:

- Churn values restricted to binary outcomes (0 or 1)
- Positive property pricing validation
- Non-negative sales values

Assertions were used to confirm logical correctness.

```python
assert sales["Total_Sales"].min() >= 0
assert churn["Churn"].isin([0,1]).all()
assert house["Price"].min() > 0

print("All tests passed")
```

✓ 0.0s Python

```
All tests passed
```

11. Business Insights

Sales Insights

- Revenue concentration suggests targeted marketing opportunities.
- Regional optimization may improve performance.

Customer Insights

- Long-term contracts improve retention.
- Pricing strategies influence churn risk.

Market Insights

- Property size is a major pricing determinant.
- Market segmentation exists across price ranges.

```python
print("===== BUSINESS SUMMARY =====")
print("Total Revenue:", total_revenue)
print("Average House Price:", avg_house_price)
print("Churn Rate:", round(churn_rate,2), "%")
```

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```
===== BUSINESS SUMMARY =====
Total Revenue: 12365048
Average House Price: 24883658.333333332
Churn Rate: 10.6 %
```

12. Business Recommendations

Based on analysis:

1. Focus promotional campaigns on high-performing regions.
2. Introduce loyalty incentives for long-term customers.
3. Encourage extended subscription contracts.
4. Apply data-driven pricing strategies in real estate decisions.
5. Monitor churn metrics continuously.

13. Implementation Plan

Phase	Action	Purpose	Timeline
Phase 1	Prepare analytics dashboard	Monitor sales, churn, and pricing metrics	2 Weeks
Phase 2	Improve marketing strategy in high-performing regions	Increase sales revenue	1 Month
Phase 3	Introduce customer loyalty offers	Reduce customer churn	1 Month
Phase 4	Apply data-based property pricing strategy	Improve pricing decisions	2 Months
Phase 5	Continuous KPI monitoring	Track business performance improvements	Ongoing

14. Limitations

- Limited dataset size for sales analysis
- No real-time data integration
- External economic factors not included

Future work may incorporate predictive modeling and machine learning.

15. Conclusion

This project successfully demonstrated an end-to-end business analytics workflow using sales, customer churn, and housing market datasets. Through data cleaning, exploratory analysis, and KPI evaluation, meaningful patterns influencing business performance were identified. The analysis showed how data-driven approaches can support decisions related to revenue improvement, customer retention, and pricing strategies. By combining visualization with structured analysis, the project highlights the practical value of data analytics in solving real-world business problems. Overall, the project reflects the application of analytical skills learned during the internship and emphasizes the importance of converting raw data into actionable insights through a reproducible workflow.